

Non-Desmos - Geometry & Trigonometry

10 original questions | Calculator permitted

Complete the questions before turning to the answer explanations. Mark any guess, even if it turns out to be correct. For math, show your setup and check the requested quantity. For reading, identify the exact supporting detail; for grammar, identify the clause or subject.

These questions were authored for this workbook from the lesson topics. They are not official SAT questions or copies of the course's paid practice.

Geometry & Trigonometry | 10 questions

[Lesson video: Geometry & Trigonometry](#)

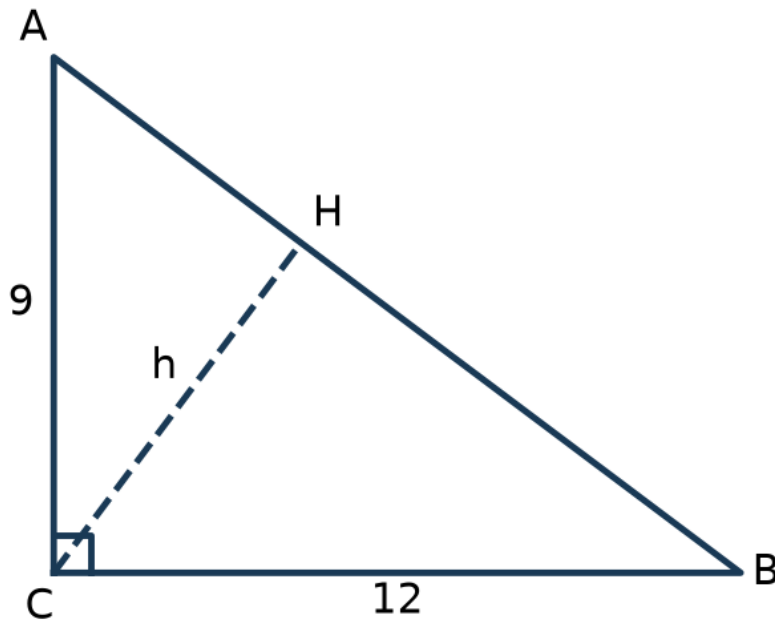
1. Lines ℓ and m are parallel. A transversal forms an angle of $(3x + 8)^\circ$ on ℓ and a same-side interior angle of $(5x + 12)^\circ$ on m . What is x ?

Answer: _____

2. In right triangle ABC, angle C is 90° and $\sin A = 5/13$. What is $\tan B$?

- A) $5/13$
- B) $5/12$
- C) $12/13$
- D) $12/5$

3. A right triangle has legs of length 9 and 12. What is the length of the altitude from its right angle to the hypotenuse?

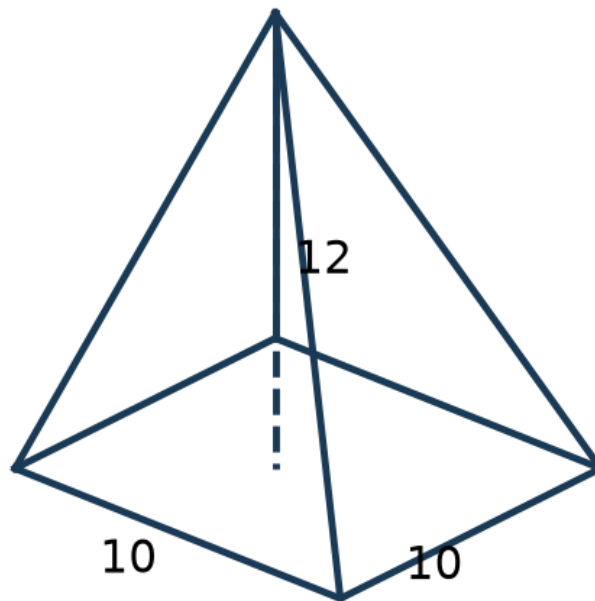


Answer: _____

4. Two similar solid containers have surface areas in the ratio 9:16. The smaller container has volume 81 cubic centimeters. What is the larger container's volume?

Answer: _____

5. A right square pyramid has base side length 10 and vertical height 12. What is its total surface area, including its base?



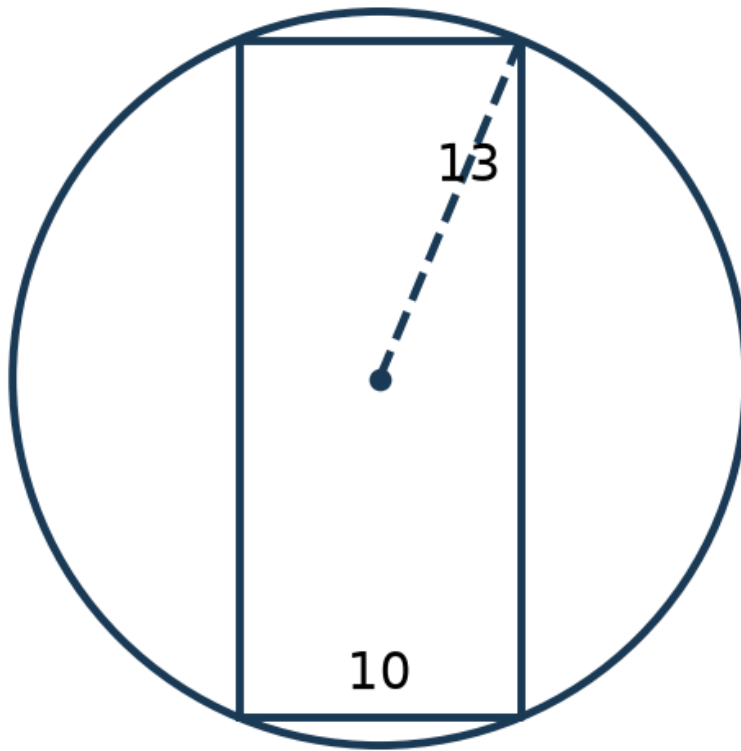
Schematic; not drawn to scale

Answer: _____

6. A cylindrical tank has base area 60 square meters and height 4 meters. A solid object occupying 35 cubic meters is placed in it. The remaining space is filled with water. What volume of water is needed, in cubic meters?

Answer: _____

7. A rectangle is inscribed in a circle of radius 13. One side of the rectangle has length 10. What is its area?



Answer: _____

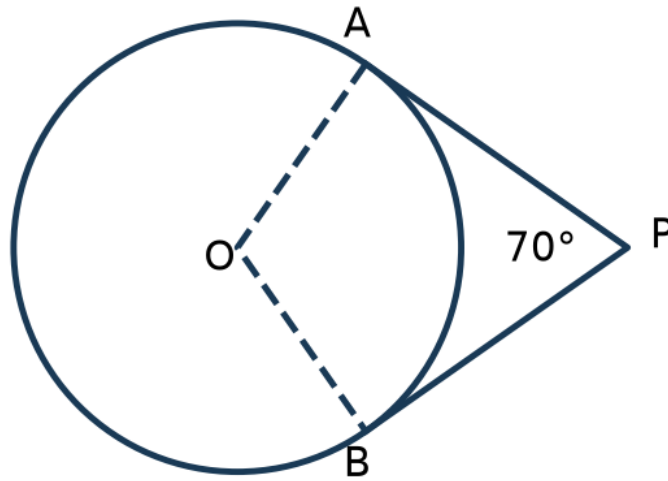
8. Triangles ABC and DEF have $\angle A = \angle D = 40^\circ$ and $\angle B = \angle E = 65^\circ$. Which additional fact guarantees that they are congruent?

- A) $\angle C = \angle F$
- B) $AB = DE$
- C) $AB = 2DE$
- D) $AC/DF = 2$

9. A triangle has two sides of lengths 6 and 11. How many integer lengths are possible for its third side?

Answer: _____

10. Two tangents from point P touch a circle at A and B. The angle APB is 70° , and the circle has circumference 72. What is the length of the minor arc AB?



Circumference = 72

Answer: _____

Answers & explanations

Count correct answers, but also review correct guesses. Redo missed questions after a day without looking at the solution.

1: 20 2: D 3: 7.2 4: 192 5: 360 6: 205 7: 240 8: B 9: 11 10: 22

Why each answer works

1. Answer: 20. Same-side interior angles between parallel lines are supplementary: $(3x + 8) + (5x + 12) = 180$. Thus $8x = 160$ and $x = 20$.

2. Answer: D. The sides opposite A, adjacent A, and the hypotenuse have ratio 5:12:13 by the Pythagorean theorem. For B, opposite is 12 and adjacent is 5, so $\tan B = 12/5$. The ratio does not specify the actual lengths.

3. Answer: 7.2. The hypotenuse is 15. Compute the area in two ways: $(1/2)(9)(12) = (1/2)(15)h$. Therefore $h = 108/15 = 7.2$.

4. Answer: 192. The length scale factor is $\sqrt{16/9} = 4/3$. The volume factor is $(4/3)^3 = 64/27$. Multiply 81 by $64/27$ to get 192.

5. Answer: 360. The slant height of a triangular face is $\sqrt{12^2 + 5^2} = 13$. Each face has area $(1/2)(10)(13) = 65$. Add four faces and the base: $4(65) + 100 = 360$.

6. Answer: 205. Tank volume = base area $\cdot$ height = $60 \cdot 4 = 240$. Subtract the solid object's 35 to get 205.

7. Answer: 240. Its diagonal is the circle's diameter, 26. The other side is $\sqrt{26^2 - 10^2} = \sqrt{576} = 24$. Area = $10 \cdot 24 = 240$.

8. Answer: B. The two equal angle pairs establish similarity. Equality of the corresponding included side AB and DE fixes the scale factor at 1 and gives ASA congruence. A adds no new information; C and D imply a different size.

9. Answer: 11. The triangle inequality requires $|11 - 6| < x < 11 + 6$, or $5 < x < 17$. The integers 6 through 16 give 11 possibilities. Both endpoints are excluded.

10. Answer: 22. Each radius is perpendicular to its tangent, so $\angle AOB = 360 - 90 - 90 - 70 = 110^\circ$. The minor arc is $(110/360)(72) = 22$.

For each missed or guessed question, record the skill, the condition you overlooked, and one specific action for next time.